

Task 1 — Inconsistencies and duplicates

CSEN1095 Data Engineering · W26 · Graded task · follows Lab 3

Deadline: Friday 9 October 2026, 11:59 PM

The situation

Student clubs collected sign-ups through an online form for one week. The file is `club_signups.csv`: 39 rows, one per form submission. Before anyone counts members or chases unpaid fees, the data needs cleaning. The same faculty, club and city are written several ways, and some sign-ups appear more than once.

Your job: turn it into a clean table with **one row per student per club**, with every value spelled one way, and explain what you did.

The data

COLUMN	MEANING
<code>signup_id</code>	The id the form gave the submission
<code>student_id</code>	The student's GUC id, e.g. 58-1234
<code>full_name</code>	As the student typed it
<code>email</code>	As the student typed it
<code>faculty</code>	The student's faculty
<code>city</code>	Where the student lives
<code>club</code>	The club they signed up for
<code>signed_up_at</code>	When they submitted the form
<code>fee_paid</code>	Whether they paid the membership fee

The canonical values. After cleaning, these columns may only contain:

COLUMN	ALLOWED VALUES
<code>faculty</code>	<code>MET</code> , <code>IET</code> , <code>EMS</code> , <code>Pharmacy</code> , <code>Management</code>
<code>club</code>	<code>Robotics</code> , <code>Debate</code> , <code>Music</code> , <code>Football</code> , <code>Chess</code>
<code>city</code>	<code>Cairo</code> , <code>Giza</code> , <code>Alexandria</code>
<code>fee_paid</code>	<code>True</code> , <code>False</code> (a boolean column, not text)

What to do

Work in a notebook, `task1.ipynb`, using pandas. Leave `club_signups.csv` unchanged, and do every fix in code.

Part A — Inconsistencies

1. **Look first.** Show the shape, the dtypes, and `value_counts()` of every categorical column, so the problems are visible before you fix them.
2. **Faculty, club, city.** Map every variant to its canonical value. Normalise mechanically first (strip spaces, lower-case), then use an explicit dictionary for what's left: `Media Engineering and Technology` is `MET`, and a `Soccer` club is the `Football` club. **Assert** that only the canonical values remain.
3. **Names and emails.** One spelling per person: no extra spaces anywhere in names, and the same capitalisation for everyone. Emails trimmed and lower-case.
4. **fee_paid.** `yes`, `Y`, `TRUE`, `1`, `no`, `N`, `false`, `0` ... become a boolean column.
5. **signed_up_at.** Two formats are mixed in this column. Find them, decide which part is the day and which the month, and parse the column into one datetime column.

Part B — Duplicates

1. **Exact duplicates.** Some rows are the same form submission twice. Count them, then remove the extra copies.
2. **The same sign-up submitted again.** Some students signed up for the same club twice, with different `signup_id`s. Usually they came back to mark the fee as paid. Keep **one row per student_id and club**: the **latest** submission. Count how many rows this removes.
3. **Check:** after Part B, the combination `student_id` + `club` must be unique. Assert it.

Save the result as `club_signups_clean.csv`.

Part C — Explain what you did

In your repository's `README.md`, write **one paragraph (about 150–250 words)** explaining your cleaning. It must cover:

- each problem you found, and how you fixed it;
- **how many rows** you started with, how many each duplicate step removed, and how many are left;
- **why** you kept the latest submission, and not the first;
- the **order** of your steps. Would you find the same duplicates if you removed them *before* fixing the `club` spellings? Say what happens, with numbers;
- two different students have the same name. How did your cleaning avoid merging them?

What to submit

A GitHub repository named:

```
{first-second}_{id}_{major}-{lab#}_task1
```

For example: `mohamed-ahmed_58-1234_MET-01_task1`

It should contain:

```
mohamed-ahmed_58-1234_MET-01_task1/
```

└─ task1.ipynb	runs top to bottom without errors, outputs visible
└─ club_signups.csv	the file you were given, unchanged
└─ club_signups_clean.csv	your result
└─ README.md	your name, ID, and the Part C paragraph

Add de-guc as a collaborator: on GitHub, open the repository → **Settings** → **Collaborators** → **Add people** → `de-guc`. The repository may be private, but without this invitation we can't see it, and it can't be graded.

Submit the repository's URL through the submission form: <https://forms.gle/TVSoNrf15Uaq93oN9>

A repository that isn't submitted through the form isn't graded, even if `de-guc` was added.

The deadline is **Friday 9 October 2026, 11:59 PM**, for both the form and the repository. The last commit before the deadline is the one graded.

How it's graded

	POINTS
Part A — Inconsistencies	
<code>faculty</code> , <code>club</code> , <code>city</code> canonical, with an assertion	20
Names and emails consistent	10
<code>fee_paid</code> is a boolean column	5
<code>signed_up_at</code> parsed into one datetime column, day and month the right way round	10
Part B — Duplicates	
Exact duplicates counted and removed	10
One row per student per club, keeping the latest, with an assertion	20
Part C — The paragraph	
Covers every point above, with correct numbers	20
Submission	
Repository named correctly, <code>de-guc</code> added, URL submitted through the form, notebook runs top to bottom	5
Total	100

Rules

- **Every fix is code.** Editing the CSV by hand, in Excel or a text editor, gets zero for that part.
- **Your own work.** You must be able to explain every line of your notebook if asked.